Site-Neutral Payments: The Effect on Long-Term Acute Care Hospitals

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Link to Latest Draft

Abstract

Medicare payment differs for similar services across care settings, creating an incentive for healthcare providers to treat patients where reimbursement is highest and unncessarily raising costs to the program. Congress has established "site-neutral" policies seeking to align patient clinical need and treatment intensity by decoupling payment from the service location. Such policies reduce cost, but could also limit access to higher-level care settings for patients who need them. I study Medicare's first site-neutral payment reform. Implemented in 2016, Long-Term Acute Care Hospitals - which provide extended hospital-level care - are reimbursed at the lower, "site-neutral" acute care hospital rate for certain low acuity patients. To evaluate the policy, I construct simulated revenue responses at the diagnosis and facility levels that predict policy-induced reimbursement changes using pre-policy patients and post-policy rules. I find that LTACHs responded to the site-neutral policy by admitting fewer low acuity patients with diagnoses facing larger reductions in payment, while admission of higher acuity patients remained constant. At the facility-level, this lead to increased closure for facilities facing greater exposure. Surviving facilities saw reductions in profitability, but were able to mitigate losses by admitting more patients with commercial insurance.

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1 Introduction

Medicare disburses over $1T annually to healtcare providers - primarily through administratively set prices (*NHE Fact Sheet | CMS* 2026). One of the most controversial elements of these prices is that Medicare payment differs for a similar set of services based on site of care. This creates a financial incentive for healthcare providers and organizations to treat patients in the highest reimbursed setting, independent of clinical necessity, and unnecessarily increasing cost to the program. Recent legislation looks to better align patient clinical need with treatment intensity by eliminating this payment differential (Cooper et al. 2023). These "site-neutral" payment policies have received significant attention as a tool to curb unnecessary Medicare spending; however, they could also negatively affect quality of care by restricting access to higher-level sites of service for patients who need them. Understanding provider responses to these policies is a crucial concern as prices are the prinicipal way regulators influence industry conduct.

An area of active debate is whether site-neutrality should be applied to post-acute care (PAC), a broad set of rehabilitative and skilled nursing services typically used following a surgery or acute medical event (Carter et al. 2016). Nearly 15% of Traditional Medicare spending is on PAC (MedPAC 2025). This covers a variety of settings and service intensities, ranging from part-time custodial services at home to extended hospital-level treatment, reflecting the continuum of patient need. However, there are limited clinical guidelines defining where patients should be treated and many patients could successfully receive care across multiple, weakly substitutable settings (Buntin 2007, Gage 2012). Moreover, disparate payment systems, both between acute care and PAC, and across PAC providers, provide minimal incentive for efficient PAC use. Nearly three quarters of unexplained geographic variation in Medicare expenditures is attributable to PAC (Newhouse et al. 2013). As a result, Medicare pays substantially different amounts for clinically similar patients based on where they receive care without seeing commensurate gains in health.

I focus on a PAC setting particularly mired in concerns about inappropriate admissions: Long-Term Acute Care Hospitals (LTACHs). LTACHs are the most intensive PAC setting and Medicare is the dominant payer: nearly 60% of admissions are for Tradtional Medicare patients (MedPAC 2014). Medicare reimburses LTACHs over seven times the acute care hospital rate, reflecting that these facilities are meant to treat patients that require prolonged hospital-level care - such as those with major organ system failure or requiring mechanical ventilation (MedPAC 2014). However, LTACHs often treat lower acuity patients to maximize profit, despite evidence that they could to

be successfully treated in lower cost settings, such as a skilled nursing facility or an acute step-down unit (MedPAC 2004). In response, as part of the Pathway for SGR Reform Act of 2013, Congress established its first site-neutral payment policy.[1] Effective 2016, LTACHs only receive the full Medicare rate for certain high acuity patients such as those who spent at least 3 days in an intensive care unit during their preceding acute care stay (*Pub. L. 113–67* 2013). Treatment for all other patients is reimbursed at the lower acute care hospital payment rate. Based on 2015 LTACH discharges, 45% of Traditional Medicare patients would have been subject to the policy, amounting to a $673 million reduction in industry revenue, 12% of total Medicare revenue.

This legislation created a strong incentive for LTACHs to reduce admissions of lower acuity patients while preserving access for higher acuity patients. However, reductions in lower acuity admissions depend in part on LTACHs' margin for such patients under the acute care hospital payment rate. Conversely, this policy could have unintended effects on access to LTACH care for higher acuity patients, for example, if the legislation had adverse financial impacts on LTACH facilities. Two natural questions arise. First, did LTACHs respond to the site-neutral policy by screening out less profitable, lower acuity patients or did they also treat fewer high acuity patients? Second, given Medicare was the dominant payer of LTACH services, how did facilities alter their operations in response to the policy? For example, did facilities close or were they able to offset operations in other ways such as adjusting their patient mix or variable inputs like staffing level? Understanding these questions is important for evaluating the overall welfare implications of the policy.

I begin by examining the effect of site-neutral price reductions on LTACH treatment patterns using the universe of LTACH admissions for Traditional Medicare patients. The primary challenge to studying the impact of this policy is that price is endogenous to the composition of patients admitted to LTACHs, which is non-random given the policy was intended to screen out lower acuity patients. To address this issue, I take a simulated revenue approach whereby I predict policy-induced reimbursement changes using pre-policy patients and post-policy rules to capture differential incentives to respond to the policy ex ante (Dafny 2005,Gross et al. 2024). Leveraging variation created at the diagnosis-level, I find that LTACHs responded to the policy by admitting

[1]Media and academic attention has been on the use of site-neutral payments in the hospital outpatient department setting beginning in 2017, often mischaracterizing it as the first and sole application of this type of policy. The details of that policy are described and evaluated in Post et al. 2025. Cooper et al. 2023 provides details on current legislative efforts to expand site-neutrality withiin this setting.

fewer patients with diagnoses facing the largest reductions in payment, consistent with a traditional upward-sloping supply curve. My point estimates imply a price elasticity of 2.3. Moreover, these results are concentrated within the set of lower acuity admissions suggesting that there was no deleterious effects on access for higher acuity patients.

Next, I estimate the effect of the site-neutral policy on LTACH-level changes by combining a similar, simulated revenue approach and detailed administrative data on LTACH operations reported annually. I find that almost 20% of facilities closed within three years after the policy was implemented, with evidence that this is weakly concentrated amongst facilities facing the greatest exposure. Surviving LTACHs saw large reductions in profitability, with Medicare margins falling the most for facilities facing the largest pre-period exposure to the policy. By contrast, profitability for commercially insured members, while falling on average, was increasing in the level of exposure. To assess this further, I investigate changes in patient mix next. I find that Medicare admissions fell on average following the policy with no changes on the margin for facilities facing greater payment reductions. This suggests that the reductions in Medicare profit margins were mechanical as facilities facing greater financial loses prior to the policy continued to do so following the policy. Interestingly, total industry discharges were unchanged. Facilities fully offset the reduction in Medicare admissions by substituting to commercially insured patients. I do not find any evidence that facilities responded by adjusting level of operation either through changes in the number of employees or beds.

My work contributes to three areas of research. First is the growing literature on site-neutral payments. To date, most attention has been on its application on the hospital outpatient departments (HOPDs). Eliminating the payment differential between HOPDs and physician offices could generate substantial static savings for Medicare and has potentially large dynamic implications for provider consolidation (Cooper et al. 2023). However, the policy initially applied to only a narrow set of locations, and estimated effects have been muted (Post et al. 2025). I build on this by evaluating a site-neutral payment policy impacting a much larger share of industry revenue. Moreover, I am the first to evaluate this in the PAC setting, an area which has received renewed interest in payment unification efforts.

Next, I contribute to the work on the treatment behavior of PAC providers. Because home health agencies and skilled nursing facilities provide most PAC, research has focused on these settings, often through the lens of wasteful spending. LTACHs have received less attention because they treat a small share of PAC patients, yet their high cost per stay and overall patient severity

makes them an important from a policy perspective. Existing LTACH research establishes two facts. Einav et al. 2023 use LTACH entry to show that most LTACH patients would otherwise have gone to SNFs, with no measurable benefit from LTACH care, and estimate that Medicare could save about $4.6 billion annually by not allowing discharge to LTACHs. Separately, Eliason et al. 2018 show that LTACHs time discharges around payment thresholds, and Einav et al. (2018) also study incentives within the LTACH payment system. However, these studies do not show how LTACHs change admissions when Medicare lowers the price of treating certain patients. I address this gap.

Finally, more broadly, my work adds to the literature examining how healthcare provider respond to financial incentives. Two papers focus on the LTACH setting. However, they provide structural estimates based on length of stay incentives within the Medicare payment structure for LTACHs. Moreover, they explicitly assume that there are no volume responses in their counterfactual analysis (Einav et al. 2018, Eliason et al. 2018). I build on this by providing the first causal estimates to price changes. As a result, my study is more comparable, both in spirit and methodology, to work by on the inpatient setting. I find an implied price elasticity of volume of 2.3, nearly double that of Gross et al. 2024 and four times that of Dafny 2005. This is consistent with LTACHs having greater discretion over admissions than acute care hospitals.

The remainder of this paper is organized into five sections. Section 2 provides background on LTACHs and the implementation of the site-neutral payment reform. Section 3 describes the data. Sections 4 and 5 present the DRG and LTACH level analyses, respectively. Section 6 discusses policy implications and concludes.

2 Background

LTACHs are the most intenstive PAC provider treating patients who require prolonged hospital-level care. This is definitional: to receive LTACH designation from Medicare, facilities must qualify as an acute care facility and have an average patient length of stay of at least 25 days (*Pub. L. 97–248* 1982). LTACH designation exempts facilites from Medicare's inpatient prospective payment system (IPPS) for acute care hospitals, instead being reimbursed under the long-term care prospective payment system (LTC-PPS).[2] Both systems pay a pre-determined, fixed rate based on clusters of

[2]LTACH designation was originally created as an exemption for about 40 facilities – primarily tuberculosis and chronic disease hospitals – specializing in patients who require prolonged inpatient treatment that were at-risk of closure during Medicare's transition to the IPPS in 1984. Liu 2001 provides a detailed history.

diagnoses referred to as diagnosis-related groups (DRGs). However, the LTC-PPS is distinct in two important ways. First, facilities are paid significantly higher within a DRG, reflecting the costs of extended care. The base Medicare rate per admission to LTACHs in 2015 was $41,043 compared to $5,437 thousand for acute care hospitals (*79 FR 49854* 2014). Second, to discourage unnecessarily brief admissions to maximize profit, facilities only receive the full LTC-PPS payment if patients stayed past a DRG-specific number of days, referred to as the short-stay outlier (SSO) threshold. Stays below the SSO are paid at the average per-diem cost (MedPAC 2014). At its peak, over 100,000 Traditional Medicare patients were treated at LTACHs annually resulting in $5 billion in program outlays (MedPAC 2018).

Prior to the policy, there were over 400 LTACHs nationwide. Unique to the industry, nearly half are "hospitals-within-hospitals" (HwH). HwHs are facilities colocated within the building or campus of a separate acute care hospital, although are required to be separately owned and operated from the hosting facility (*42 CFR 412.22* n.d.). Nearly three-quarters of LTACHs are for-profit and highly profitable. Industry-level operating margins for Medicare patients, which account for over 60% of LTACH admissions, are frequently near double digits (MedPAC 2014).[3]

Long-term acute care hospitals (LTACHs) treat medically complex patients who remain too ill or unstable for skilled nursing or intensive rehabilitative services, but who no longer need the diagnostic and procedural intensity of a general acute care hospital. Care often includes daily physician oversight, round-the-clock nursing, respiratory therapy, and multidisciplinary management of multiple conditions (Gage et al. 2005. Common clinical needs include ongoing management of respiratory failure, treatment of severe infections or sepsis requiring long-term intravenous antibiotics, complex wound care, and support for patients with multiple organ failure. Patients frequently arrive following a short-term acute care hospitalization, including 35% who are transferred from an intensive care unit, and approximately 15% require prolonged mechanical ventilation (MedPAC 2014,MedPAC 2018). The 30-day mortality rate for LTACH patients is 24%, underscoring the overall severity of illness treated at these facilities (MedPAC 2014).

However, there is significant overlap in the types of patients treated at LTACHs and in other PAC settings. For example, while musculoskeletal aftercare was the second most common LTACH diagnosis group in 2004, 80% of Medicare patients with these diagnoses were discharged to skilled

[3]The two largest LTACH operators, Kindred Health Systems and Select Medical Holdings, reported profit margins ranging from 16% to 21% prior to the policy. Profit margins based on earnings before interest, taxes, depreciation, and amortization. Data obtained from annual investor reports (10-K) submitted to the SEC.

nursing facilities (Kennell and Associates 2010). Even patients requiring mechanical ventilation, seemingly the high-severity patients most likely to benefit from LTACH care, are also treated in less costly settings, such as SNFs or acute care step-down units, with no differences in 90-day mortality rates. This overlap reflects the limited evidence-based clinical guidelines specifying which patients are most appropriate for LTACH care. While clinicians agree that LTACH treatment should be reserved for the chronically, critically ill," there is no consensus definition of this population (Kennell and Associates 2010). Instead, transfer decisions to LTACHs largely reflect non-clinical factors, such as supply availability and physician preference (Kennell and Associates 2010). Consequently, LTACHs and lower-acuity PAC providers treat patients with substantially overlapping clinical profiles, with no differences in 90-day mortality or improvements in other observable measures of quality at LTACHs. Given the three-fold difference in Medicare price per stay, $36,000 at LTACHs and $12,000 at SNFs, this overlap could be a substantial source of wasteful spending (MedPAC 2018). This led regulators to question the purpose of these facilities, including statements such as "The lack of clear evidence on costs raises questions about the value of Medicare expenditures on LTCH care" (MedPAC 2019).[4]

Admission of lower-acuity patients is financially advantageous for LTACHs. Fixed-price reimbursement under the LTC-PPS incentivizes the treatment of lower-cost patients within a DRG, who tend to be lower acuity, in order to maximize profit. This behavior is exacerbated by acute care hospitals facing similar incentives under the IPPS. While acute care facilities have less discretion over which patients are admitted, they may adjust treatment intensity, including by quickly transferring patients following stabilization. Together, these incentives may encourage facilities to ping-pong patients between acute care hospitals and LTACHs to maximize profit, with each new admission generating additional Medicare payment. CMS attempted to limit this behavior by penalizing LTACHs which either receive a large portion of admissions from a single acute care facility or whose patients are frequently readmitted to an acute care facility.[5] Unfortunately, these efforts were largely ineffective because they did not target the underlying issue: it is unclear which patients should be at LTACHs.

In response, a major payment reform looking to target unnecessary, low acuity admissions was

[4]One policy brief goes further to suggest that elimination of the designation altogether would immediately save Medicare money without any deleterious effects on health (`https://www.onepercentsteps.com/policy-briefs/long-term-care-hospitals-a-case-study-in-waste`).

[5]These policies are referred to as the 25% rule and the 5% rule, respectively. Einav et al. 2023 describes these policy changes in greater detail.

announced as part of the Bipartisan Budget Act of 2013 (*Pub. L. 113–67* 2013). Beginning in 2016, LTACHs only receive the standard LTC-PPS rate for patients meeting all the following criteria:

1. Transferred directly from an acute care hospital
2. Not admitted for one of 13 psychiatric/rehabilitation DRGs
3. Either spent at least three days in an intensive care unit (ICU) or coronary care unit (CCU) during their preceding stay; **or** required at least 96 hours of mechanical ventilation weaning during their LTACH stay [6]

Treatment for all other patients is reimbursed at a lower, "site-neutral" IPPS rate for the same DRG. The reform was phased-in between 2016 and 2019 at the 50-50 average of the LTC-PPS and the IPPS rates. It was intended to take full effect in 2020, but was delayed for the duration of the COVID-19 Public Health Emergency, taking effect in 2023 (MedPAC 2021). This payment reform led to large changes in reimbursement rates. For example, the 2015 baseline Medicare rate for a patient with pulmonary edema – DRG 189 – was $37,341 under the LTC-PPS and $6,599 under the IPPS.

2.1 Hypotheses

Site-neutral payment reform substantially reduced Medicare reimbursement for affected LTACH admissions. Moreover, because Medicare was such a dominant payer of LTACH services, this policy change lead to a large reduction in industry revenues as well. I therefore consider responses at the DRG and LTACH levels.

DRG-level responses

The policy was intended to financially disincentivize the treatment of low acuity patients, making patient volume responses a first order concern. I therefore hypothesize that admissions of site-neutral patients will decline, particularly in DRGs with greater exposure to the policy. These declines may also spill over to unaffected patients if facilities respond to the policy by reducing their overall operations.

The site-neutral payment policy may also affect patient length of stay because of its connection to LTC-PPS reimbursement through the SSO threshold. Prior research shows that LTACHs are

[6] This was less restrictive than MedPAC's recommendation of at least 8 ICU days (MedPAC 2014). Based on my calculation of 2013 LTACH discharges, only 37% of admissions qualify under the 8 ICU day criteria compared to 57% under the current policy.

sensitive to the short-stay outlier (SSO) threshold, with discharges concentrated around the point at which payment increases substantially (Kim et al. 2015, Einav et al. 2018, Eliason et al. 2018). However, the direction of the change is not immediately clear. For example, if the policy entirely screened out site-neutral patients, then we may expect the average length of stay to increase because only higher acuity members are still treated at LTACHs. Alternatively, facilities may differentiate within site-neutral admissions and only accept the most profitable post-policy change. I treat this as an empirical question.

LTACH-level responses

The policy reduced the marginal reimbursement of LTACHs' most profitable payer, Medicare. As such, I expect a negative effect on Medicare profit margins. Medicare often assesses the adequacy of its payments using provider margins, making them a policy-relevant measure of the financial consequences of site-neutral payment. Commercial payers often use Traditional Medicare payment policy as the basis for rate negotiations with providers. I therefore expect that that commercial margins will fall as well.

Lower Medicare payment rates increase the relative value of treating patients with other types of insurance (McGuire et al. 1991). Facilities may alter their mix of patients from those covered by Traditional Medicare to other payers to offset losses. Shifting towards non-Medicare patients, such as Medicare Advantage and commerical payers, was a key strategy reported by two of the largest for-profit LTACH operators leading up to the implementation of the site-neutral policy (Select Medical Holdings 2015,Kindred Health 2014). I hypothesize that the number of Medicare admissions will decrease and the numer the non-Medicare admissions will increase.

Finally, given the magnitude of the payment reduction, some LTACHs may be unable to offset lost revenue through changes in patient mix or length of stay alone. I therefore expect facilities to reduce their scale of operations, which may ultimately result in facility closure.

3 Data

I rely on two primary datasets. First, I use the Medicare Limited Dataset (LDS) Inpatient Claims Standard Analytic Files for calendar years 2010-2019. These data include all inpatient insurance claims submitted to Medicare for fee-for-service beneficiaries, which I group into inpatient discharges.[7] Each discharge has the following information: beneficiary identifier, hospital identifier,

[7]Less than 1% of total or LTACH hospitalizations are associated with multiple claims. I consolidate these these into unique hospital discharges using a modified version of the logic provided by the Research Data Assistance

DRG, revenue and procedure codes, dates of service, and Medicare reimbursement amounts. Starting in 2016, discharges also include an indicator for whether they were paid under the site-neutral logic. Hospitals are identified via their CMS Certification Number (CCN), which I use to identify LTACH discharges.[8]

Second, I use the Hospital Cost Report Information System (Cost Reports) for fiscal years 2011-2018. Fiscal years are accounting constructs that can begin anywhere in the year and often span multiple calendar years. I choose these Cost Report fiscal years because they are completely uncompassed by the set of calendar years in the claims data. The Cost Reports contain facility-year level financial and operational information for all providers participating in the Medicare program. This includes data on hospital identifiers, revenue and cost by type of payer (Medicare, Medicaid, other), discharges by type of payer (Medicare, Medicaid, and other) as well as high-level operational metrics such as number of full-time equivalent employees (FTEs), total salaries, and number of beds. LTACHs are similarly identified via CCN. I use this to link to the discharge data and use cost-reporting dates to subset to discharges to within the same fiscal years.

Table 1 presents descriptive statistics for the entire study period by type of hospitalization. As expected, LTACHs care for a relatively more severe set of patients within the distribution of all hospitalizations. This is reflected across multiple measures of intensity including the mean IPPS DRG weight (1.6 vs 2.5), the mean number of ICU days (1.6 vs 8), and the percentage of individuals requiring mechanical ventilation (3.9% vs 20.3%). However, there is significant variation when decomposing by site-neutral status. Site-neutral admissions are more like the average hospitalization, meaning that differences in aggregate are largely driven by the acuity of non-site neutral patients.

Center (ResDAC) (`https://resdac.org/articles/medicare-inpatient-file-lds-or-rif-counting-distinct-hospitalizations`).

[8]CCNs ending in 2000-2299 are defined as "Long-Term Acute Care Hospitals (Excluded from PPS)".

Table 1: Descriptive Statistics

	All Hospital Discharges	LTACH Discharges		
		Total	Standard	Site Neutral
Total Discharges	110,234,558	1,010,340	588,900	421,440
Panel A: Patient Demographics				
Age (yr)	71.9	70.0	70.3	69.6
Male (%)	45.4	51.0	51.9	49.8
Dual Eligible (%)	27.9	38.6	38.1	39.3
Race/Ethnicity (%)				
White	81.0	71.8	72.5	70.8
Black	12.8	20.2	19.0	21.9
Hispanic	2.2	3.6	3.6	3.6
Asian	1.3	1.6	2.0	1.1
American Indian/Alaska Native	0.7	0.9	0.8	0.9
Other	1.3	1.3	1.4	1.1
Panel B: Clinical Characteristics				
IPPS DRG Weight	1.6	2.5	2.9	1.8
Any ICU, Prior Acute Stay (%)	35.8†	65.9	99.1	19.6
ICU Days, Prior Acute Stay	1.6†	8.0	13.0	0.8
Any Mechanical Ventilation (%)	3.9	20.3	32.3	3.6
Panel C: Utilization and Payment				
Length of Stay (days)	6.8	26.7	27.7	25.2
Mean Medicare Payment ($)	11,651	37,203	42,992	29,115
Panel D: Discharge Destination (%)				
Home	45.7	8.9	6.1	12.8
Home Health Agency	17.6	19.3	15.0	25.5
Skilled Nursing Facility	19.7	36.1	37.4	34.3
Inpatient Rehabilitation Facility	3.5	6.0	7.0	4.5
Long-Term Acute Care Hospital	1.1	0.4	0.4	0.3
Died	3.2	12.4	16.2	7.0
Other	9.3	17.0	18.0	15.6

Note: Sample is Medicare fee-for-service discharges for calendar years 2011-2019. Standard LTACH discharges are those meeting the site neutral qualifying criteria and paid the full LTC-PPS rate. For LTACH discharges, ICU measures are calculated from the acute care stay preceding the LTACH admission if any. †For all hospital discharges, ICU measures are based on ICU use during the hospitalization itself. Payments are nominal means per discharge. Discharge destination is based on discharge status codes.

4 DRG-Analysis

I begin by evaluating the effects of diagnosis-specific price reductions resulting from the policy on LTACH treatment patterns. A well-known challenge in estimating the casual impact of price in this context is that price is endogenous to the composition of admitted patients. For example, if

providers respond to the policy by reducing site-neutral patient volume, then the average Medicare price will increase mechanically. This will introduce spurious negative correlation to a direction regression of price on my outcome of interest. As such, I follow existing literature and estimate the effect of price changes using a simulated revenue approach which isolates the mechanical effect of the policy from the behavioral response by simulating the effect it would have had on observations prior to its implementation (Dafny 2005,Gross et al. 2024).

4.1 Simulated Revenue Definition

My explanatory variable of interest is the average predicted per discharge exposure to the site-neutral payment reform. I calculate this by simulating the price under the site-neutral payment system for all LTACH discharges in 2013, the year the policy was announced. This is a two-step process. First, I identify discharges that would have been subject to the site-neutral rate using the qualifying criteria.[9] Second, I apply the site-neutral pricing logic using Medicare rule-making data and provider-specific reimbursement data maintained by Medicare Administrative Contractors that adjudicate Medicare fee-for-service claims.[10] Both sources are publicly available on the Federal Register and CMS website, respectively. Each step is validated using post-period discharges. Appendix Figure 7 presents a bin-scatter relating my simulated price to the actual Medicare price by ventile. Regression results produce slope and R^2 parameters near unity.

Next, I calculate the average simulated revenue change as the difference between my simulated price and the actual Medicare allowed amount at the DRG level:

$$shock_{d,2013} = \frac{1}{N_{d,2013}} \sum_{\{i:d(i)=d\}} (\widehat{Price}_{i,2013} - Price_{i,2013}) \tag{1}$$

Where $Price_{i,2013}$ is the actual Medicare price, $\widehat{Price}_{i,2013}$ is the predicted payment under the site-neutral policy, and $d(i)$ is the DRG for discharge i. The variable $shock_{d,2013}$ represents the

[9]Revenue codes for ICU days are 200-214. Coronary Care Unit stays also qualify and are billed using revenue code 219. Mechanical ventilation for at least 96 hours is identified by ICD-9 codes 96.70 and 96.72, and ICD-10 code 5A1955Z. Psychiatric and rehabilitation DRGs are 876, 880, 881, 882, 883, 884, 885, 887, 894, 895, 896, 897, 945, 946.

[10]For exceptionally expensive patients, Medicare pays facilities an additional amount above the fixed PPS rate to defray the cost of treatment. This is referred to as high-cost outlier (HCO) payment. HCO payment covers 80% of costs above a pre-determined cost threshold and applies to 15% of LTACH discharges annually. I do not incorporate this into my simulated pricing logic under the assumption that facilities are only minimally responsive to this aspect of payment given their limited scope.

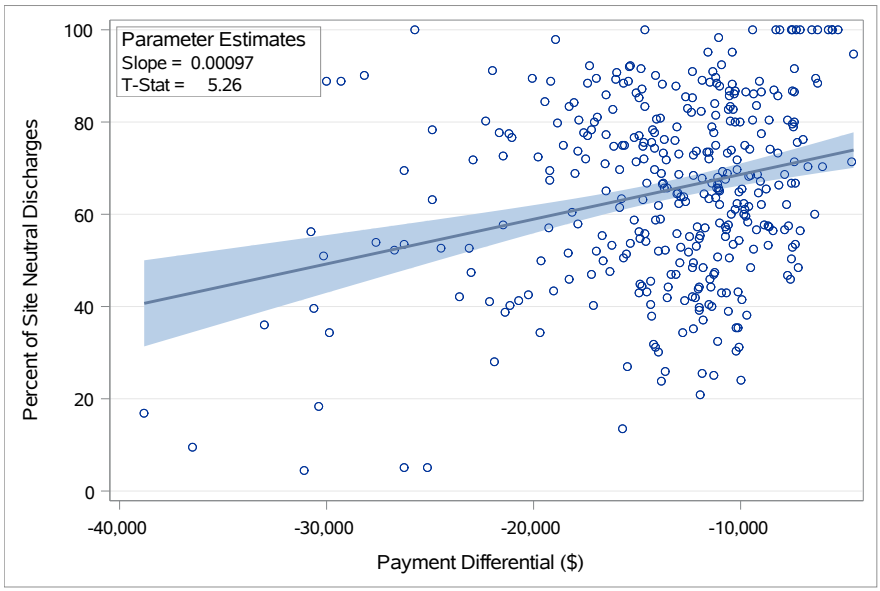
DRG-specific average per-discharge revenue reduction had facilities done nothing. Importantly, this also captures the differential incentive to respond to the policy and is my regressor of interest.[11]

This combines two sources of variation into a single Intent-to-Treat exposure. The first source of variation is in the spread between the LTC-PPS and IPPS rates. For example, in 2013, the differential between these two systems for a patient with kidney or urinary tract infection (DRG 610) was $16,153 compared to $35,164 for patient with skin debridement (DRG 919). The second source of variation is in the proportion of admissions qualifying for site-neutral payment. This can range from near zero, for DRGs defined based on 96+ hours of mechanical ventilation (DRG 207), to 100% for the 13 psychiatric and rehabilitation DRGs. Figure 1a presents the bivariate relationship between these two sources of variation. Figure 1b plots the distribution of my regressor of interest: the simulated revenue change across DRGs. The average simulated revenue change is -$6,076 per discharge (SD: $3,013), a 17% reduction. Appendix 6.2 explores an alternative specification.

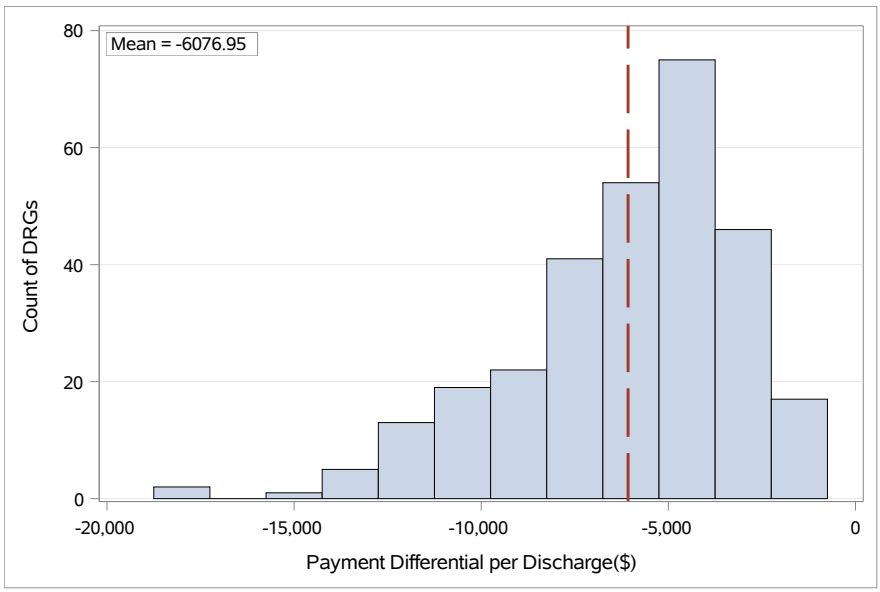
[11]To ensure that my estimates are based on services that are consistently provided by LTACHs, I only consider DRGs with at least 30 LTACH discharges from between 2011-2013. I choose this cut-off because Medicare only defines relative payment amounts (referred to as weights) for DRGs with at least 10 discharges within a fiscal year. DRGs with fewer than 10 discharges are either assigned weights based on similar clinical characteristics (1-9 discharges) or do not receive one.

Figure 1: Determinants and Distribution of DRG-Level Simulated Revenue Change

(a) Bivariate Relationship of Determinants of Simulated Revenue Change



(b) Distribution of DRG-Level Simulated Revenue Change



Note: These figures show the source of variation summarized in my DRG-level exposure of interest as well as it's distribution. Panel A is a scatterplot relating the payment differential to the percentage of site neutral discharges within a DRG. Each dot represents a DRG and the line is an OLS regression of the price differential on the percentage of site neutral discharges. Panel B is a histogram of the average simulated per-discharge revenue shock under the site-neutral policy, calculated as the average difference between the simulated Medicare price and the actual Medicare price within a DRG. Both panels are based on 2013 Medicare fee-for-service discharges.

4.2 Outcome Variable Definitions

My first outcome of interest is the volume of patients treated. For each DRG in each year, I measure volume as the number of discharges. My second outcome is length of stay, measured based on the admission and discharge dates. Both are calculated for all discharges and separately for site-neutral and non-site neutral discharges, applying the same logic used in creating the simulated revenue change.

4.3 Empirical Strategy

I model LTACH treatment decisions as a function of predicted financial losses under the site-neutral payment policy ($shock_{d,2013}$) using a continuous difference-in-differences framework that leverages variation in simulated revenue changes across DRGs. Specifically, I compare changes in outcomes for DRGs facing greater reductions in price to those of DRGs facing smaller reductions (first difference) before and after the policy (second difference) by estimating equations of the following form:

$$Y_{d,t} = \mu_d + \gamma_t + \sum_{t=2011}^{2018} \beta_t (shock_{d,2013} * \gamma_t) + X_{d,t} + \epsilon_{d,t} \tag{2}$$

Where $Y_{d,t}$ is outcome Y for DRG d at time t. The μ_d and γ_t parameters are DRG and year fixed effects, respectively. My coefficients of interest are the β_ts which estimate the effect of the DRG-specific simulated revenue shock ($shock_{d,2013}$) for each year relative to 2013. These are scaled by -$1,000 for interpretability and thus capture the change in the outcome variable for a $1,000 decrease in price. DRGs are priced based on underlying relative value (referred to as weights), which are updated annually. The $X_{d,t}$ vector includes time-varying LTC-PPS and IPPS DRG weights to control for price changes resulting from recalibration and not my policy of interest. The inclusion of these covariates is explored further in Appendix 6.1. Regressions are weighted by the number of pre-period discharges and standard errors are clustered at the DRG-level. I supplement this with regressions of the following form to estimate a single summary treatment effect:

$$\begin{aligned} Y_{d,t} = \mu_d + \gamma_t + \beta_{interim} \left(shock_{d,2013} \times \mathbf{1}[2014 \leq t \leq 2015]\right) \\ + \beta_{post} \left(shock_{d,2013} \times \mathbf{1}[2016 \leq t \leq 2018]\right) + X_{d,t} + \epsilon_{d,t} \end{aligned} \tag{3}$$

Where $\beta_{interim}$ captures anticipatory responses between the announcement and implementation of the policy, and β_{post} captures the post-period effect as single point estimates. Results are similarly

weighted by the number of pre-period discharges, scaled by -$1,000 for interpretability, and standard errors are clustered at the DRG-level.

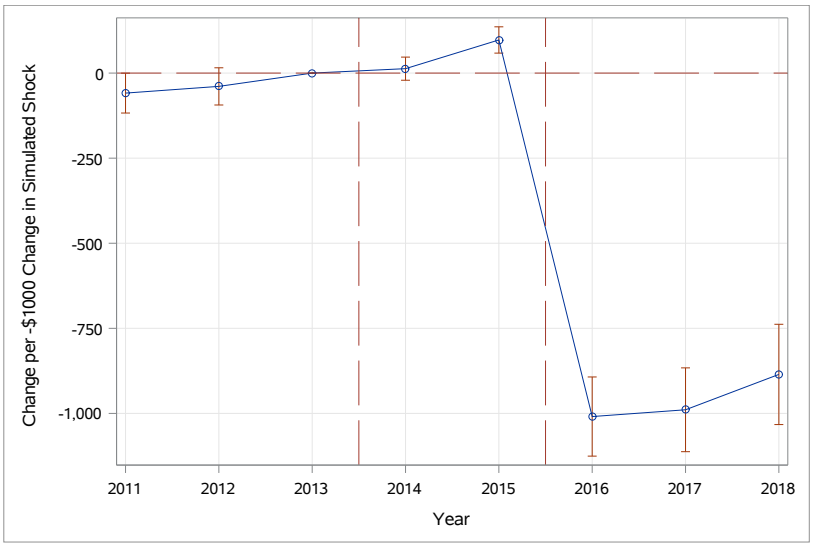
To interpret my estimates as causal, I make two assumptions. First is the traditional parallel trends assumption common to difference-in-differences frameworks. That is, changes in outcomes for DRGs across level of exposure would have developed similarly had the site-neutral policy not been implemented. I assess the plausibility of this assumption by testing for pre-period deviations in trend.

Second is the "strong" parallel trends assumption unique to the continuous difference-in-differences setting (Callaway et al. 2024). Specifically, I must assume that there is no selection on treatment gains: DRGs facing larger simulated price changes are no more responsive to reimbursement shocks than DRGs facing smaller ones. A potential violation of this assumption could arise if there is correlation between exposure to the policy and the ability to screen out site-neutral patients. For example, patients with psychiatric and rehabilitation DRGs are observable prior to admission and are site-neutral with certainty. In contrast, patients requiring mechanical ventilation are only determined to be site-neutral after admission, conditional on time spent on the machine. In Appendix 6.3, I drop psychiatric and rehabilitation DRGs as a sensitivity analysis to show they do not drive my results.

A final issue for identification is whether to pursue an instrumental variables strategy, which would estimate the local average treatment effect of the policy. Certain DRGs are completely screened out post-policy and thus are not observed throughout the entire study period. For example, prior to 2016, LTACHs treated around 300 patients annually for DRG 895 – alcohol, drug abuse, or dependence with rehabilitation therapy – and zero afterwards. This precludes unbiased estimation of a two-stage least squares instrumental variables approach because those observations will be missing Medicare price in the post period. Instead, I estimate and discuss reduced form results.

4.4 Results

Figure 2: Effect of Predicted Medicare Price on Actual Medicare Price



Note: This figure plots the βs of equation (2) using actual Medicare price as an outcome, scaling values by -$1,000 for interpretability. Year fixed effects are shown in Appendix Table 4.

Figure 2 presents my DRG-level first stage estimates of the effect of my simluated revenue change on actual Medicare reimbursement on simulated revenue shock. Each $1,000 reduction in simulated revenue is associated with a $1,021 reduction in actual Medicare price during the post-period (β = -1,020.8, $p < 0.01$). The event-study estimates show statistically significant but economically small deviations before implementation. The first-stage effect is near one-to-one in 2016 (-$1,009) and 2017 (-$989) and attenuates modestly in 2018 (-$885)

Figure 3 presents the reduced-form effects of the policy on the number of discharges and length of stay. LTACHs reduced admissions more in DRGs facing larger payment reductions (Panel A). Each $1,000 reduction in simulated revenue is associated with a 6.0% decline in discharges during the post-period (β = -0.062, $p < 0.01$). At the mean simulated revenue change, this implies a 31.4% decline in discharges and a price elasticity of volume of 2.29. The event-study estimates show no significant pre-period deviations, and the response grows over time, from -0.045 in 2016 to -0.058 in 2017 and -0.084 in 2018. At the mean simulated revenue change, these estimates imply declines of 23.9%, 29.7%, and 40.0%, respectively.

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There is also a small reduction in patient length of stay (Panel D). Specifically, the mean simulated price change implies a 6.1% reduction in length of stay or 1.59 days. However, this should be interpreted with caution due to the changing composition of patients. On the one hand, if the policy worked perfectly, then only the highest severity patients would still be treated at LTACHs. This would suggest that the average expected LOS would increase and that the change reflects a true shift in treatment. Alternatively, highly profit-motivated LTACHs could choose to only admit less severe, site-neutral patients leading to the same LOS change. Determining the source is dependent on the underlying, unobservable severity of patients over time.

Figure 3: Effect of the Policy on Number of Discharges and Length of Stay



Note: Plots the βs of equation (2) using the log number of discharges and log length of stay as outcomes. Rows group dependent variables (discharges or length of stay), columns group types of admissions (all discharges, site-neutral discharges, or non-site-neutral discharges). Year fixed effects are shown in Appendix Table 4. All values have been scaled by -1,000 for interpretability.

5 LTACH-Analysis

Next, I evaluate the effects of the site-neutral payment reductions at the LTACH-level on facility operations. This analysis faces similar endogeneity concerns as the DRG-level analysis, but mediated through operations. For example, if facilities respond to the policy by reducing the number of site-neutral admissions, Medicare price will decrease. To the extent that the reduction in admissions covaries with contractions in facility meaures, this will again introduce a spurious negative correlation. Therefore, I take a simulated revenue approach here as well.

5.1 Simulated Revenue Definition

My explanatory variable is interest is the average simulated revenue change per discharge calculated as the difference between my simulated price and the actual Medicare allowed amount at the LTACH-level:

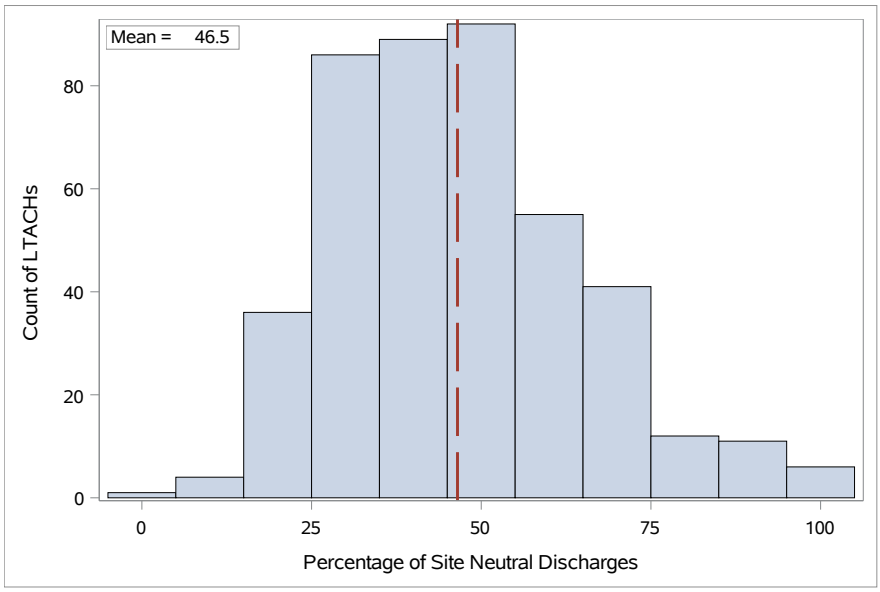
$$shock_{h,2013} = \frac{1}{N_{h,2013}} \sum_{\{i:h(i)=h\}} (\widehat{Price}_{i,2013} - Price_{i,2013}) \tag{4}$$

Where $Price_{i,2013}$ is the actual Medicare price, $\widehat{Price}_{i,2013}$ is the predicted payment under site neutral policy, and $h(i)$ is the LTACH for discharge i. The variable $shock_{h,2013}$ represents the LTACH-specific average per-discharge revenue reduction had facilities done nothing in response to the site-neutral policy.

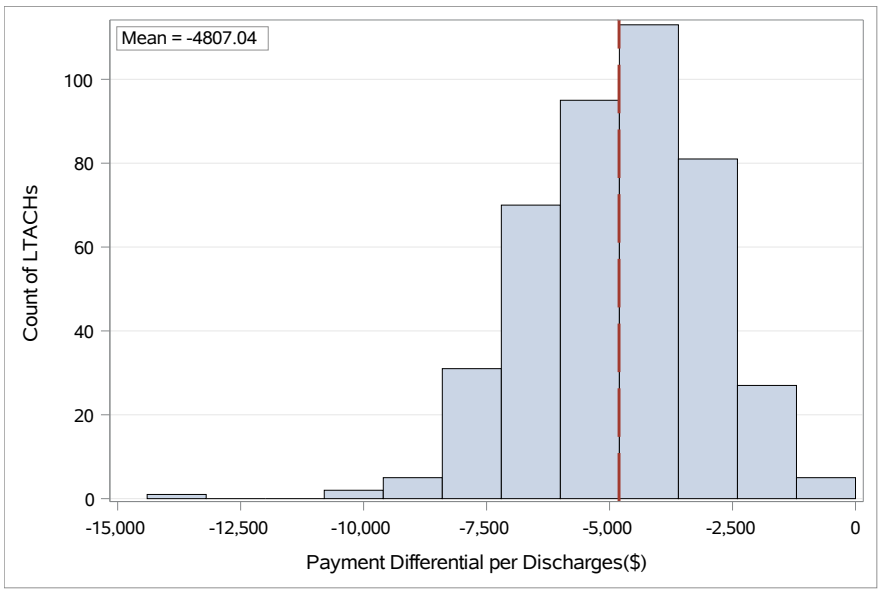
This measure summarizes the same variation in the DRG-level analysis along with variation in the percentage of site-neutral discharges across facilities. The median percentage of site-neutral discharges at the facility-level in 2013 was 42% with an interquartile range of 25 percentage point (Figure 4a). Figure 4b plots the distribution of my simulated revenue change across LTACHs. The average change in simulated revenue is -$4,807 per discharge (SD 1,751), a 13% reduction. This is smaller than the mean value at the DRG level reflecting the fact that it incorporates differences both in the number and composition of DRGs across facilities.

Figure 4: Determinants and Distribution of LTACH-Level Simulated Revenue Change

(a) Distribution of the Percent of Site-Neutral Discharges



(b) Distribution of LTACH Simulated Revenue Change



Note: Panel A is a histogram of the percentage of site-neutral discharges at the LTACH-level. Panel B is the average simulated per-discharge revenue shock under the site neutral policy, calculated as the average difference between the simulated Medicare price and the actual Medicare price for 2013 discharges within an LTACH.

5.2 Outcome Variable Definitions

My first outcome of interest is facility closure. I measure closure as the first period that a hospital identifier, CCN, stops reporting in the the Cost Reports. A known limitation of the Cost Reports is that CCNs can disppear over time for reasons other than facility closure. Hospital mergers or conversion to a new facility type may result in the discontinuation of the previous CCN.[12] I validate this measure using additional data sources such as the CMS provider of service files.

My second outcome of interest is LTACH profitability. I measure this for every LTACH in every year as operating margin, the ratio of facility net to gross patient revenue. This measure is used by CMS when evaluating the adequacy of their provider payments. Facilities report cost and revenue in total and separately for Traditional Medicare discharges and Medicaid discharges. Using this information, I construct operating margin for all discharges, Traditional Medicare discharges, and non-Traditional Medicare, non-Medicaid discharges. This last measure is intended to capture Medicare Advantage and commercial members, thus I refer to it as commercial margin. I do not evaluate Medicaid margin separately because few facilities reported treating patients with Medicaid as the primary payer prior to the policy.

My third outcome of interest is patient mix. I measure this directly from the Cost Reports as the number of discharges. Similar to revenue and costs, facilities report patient volume separately for Traditional Medicare and Medicaid covered patients. Using this information, I construct the number of discharges in total, for Traditional Medicare patients, and for non-Medicare, non-Medicaid members (commercial).

My fourth outcomes are measures of varible inputs to operation. I measure these directly Cost Reports as the number of full-time equivalent staff, total salaries, and the number of facility beds.

Reporting descrepancies are common in the Cost Reports. For example, facility cost reports may cover significantly more or less than 365 days when facilities are either opening, closing, or changing their reporting periods. I account for these outliers by winsorizing reported values above the 99th percentile for all continuous measures.

[12]The latter is the case with LTACHs. Facilities converting to LTACHs are initially paid under the IPPS using a standard acute care CCN. After meeting the minimum 25 day average length of stay requirement for a six-month period and maintaining it, facilties are officially designated as LTACHs are receive a new CCN.

5.3 Empirical Strategy

I model LTACH operations as a function of predicted financial losses under the site-neutral payment policy ($shock_{h,2013}$) using a continuous difference-in-differences framework that leverages variation in simulated revenue changes across facilities. Specifically, I compare changes in outcomes for LTACHs facing greater reductions in price to those of LTACHs facing smaller reductions (first difference) before and after the policy (second difference) by estimating equations of the following form:

$$Y_{h,t} = \mu_h + \gamma_t + \sum_{t=2011}^{2018} \beta_t (shock_{h,2013} * \gamma_t) + X_{h,t} + \epsilon_{h,t} \tag{5}$$

Where $Y_{h,t}$ is outcome Y for LTACH h at time t. The μ_h and γ_t parameters are LTACH and year fixed effects, respectively. My coefficients of interest are the β_ts which estimate the effect of the LTACH-specific simulated revenue shock ($shock_{h,2013}$) for each year relative to 2013. These are scaled by -$1,000 for interpretability and thus capture the change in the outcome variable for a $1,000 decrease in price. The $X_{h,t}$ vector includes time-varying LTC-PPS and IPPS DRG weights. Standard errors are clustered at the facility-level. I supplement this with regressions of the following form to estimate a single summary treatment effect:

$$\begin{aligned} Y_{h,t} = \mu_h + \gamma_t + \beta_{interim} \left(shock_{h,2013} \times \mathbf{1}[2014 \leq t \leq 2015]\right) \\ + \beta_{post} \left(shock_{h,2013} \times \mathbf{1}[2016 \leq t \leq 2018]\right) + X_{h,t} + \epsilon_{h,t} \end{aligned} \tag{6}$$

Where $\beta_{interim}$ captures anticipatory responses between the announcement and implementation of the policy, and β_{post} captures the post-period effect as single point estimates. Results are similarly scaled by -$1,000 for interpretability and standard errors are clustered at the facility-level.

I deviate from the prior analysis in two important ways. First, I include LTACH fixed effects for all outcomes except for closure. I do this to approximate a hazard model in a linear setting. Second, I respecify my LTC-PPS and IPPS ($X_{h,t}$) controls. Mean weights based on contemporaneous discharges are endogenous to the policy because, as shown in the DRG analysis, the policy-induced compositional changes in patients treated at these facilities. Instead, I reestimate these controls using a constant cohort of 2013 discharges where I reassign the weights for each fiscal year.

To interpret my results as causal, I must make parallel trends assumptions similar to my DRG-level analysis. Specifically, I first assume that changes in outcomes for LTACHs facing different

levels of exposure would have developed in parallel in the absense of the site-neutral policy. I again test for pre-period deviation in trend to assess the plausibility of this assumption.

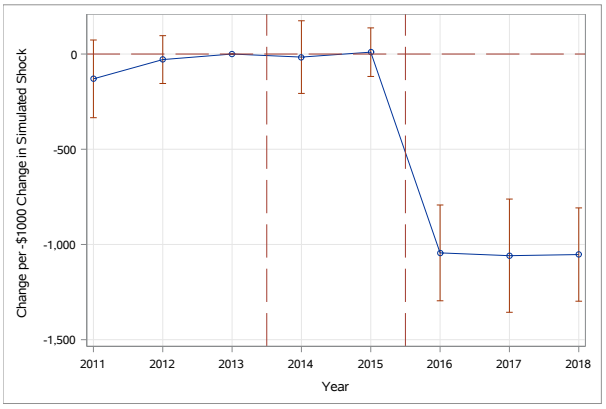
Next, the strong parallel trends assumption implies that there is no relationship between an LTACHs level of exposure to the policy and their ability to alter facility operations. A potential violation to this is assumption is if facility operations are specialized in the treatment of site-neutral patients specifically, for example, psychiatric or rehability patients. This is likely the case given the small mass of facilities treating 100% site-neutral patients shown in Figure 4a. Prior work corroborates this, finding that a small number of facilities solely treat psychiatric DRGs, despite having LTACH designation (Liu 2001). However, this same work defined a facility as specializing in mental health if at least half of their cases were for psychiatric DRGs, highlighting the fact that identifying these facilities is both difficult and arbitrary. For this reason, I include these facilities in my analyses.

5.4 Results

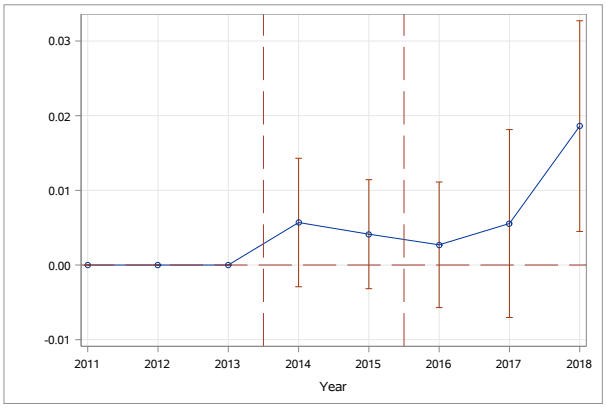
I begin by assessing the validity of my simulated revenue change. The first stage estimates presented in Figure 5a show no deviations in the pre-period, again, followed by a reduction near unity in payment after the policy implementation (DiD: -$1,049 per every -$1,000 change in predicted revenue).

Figure 5: Effect of Simulated Revenue Change on Actual Medicare Price and Closure

(a) Effect on Actual Medicare Price



(b) Effect of Facility Closure



Note: This figure plots the βs of equation (2) using the actual Medicare price (Panel A) and closure (Panel B) as an outcomes. Year fixed effects are shown in Table 3. All values have been scaled by -1,000 for interpretability.

Table 2: Characteristics of LTACHs by Closure Status

	Never Closed	Ever Closed	Difference	p-value
N	342	84		
Margin				
Medicare Price ($)	37,169	34,002	−3,167	<0.001
Operating Margin	0.046	−0.017	−0.063	0.001
Medicare Margin	0.063	0.035	−0.028	0.095
Commercial Margin	0.029	−0.094	−0.123	0.002
Discharges				
Total Discharges	494.8	366.7	−128.1	<0.001
Medicare Discharges	319.1	247.0	−72.1	0.001
Commercial Discharges	135.2	91.6	−43.6	<0.001
Operations				
For-Profit (%)	64.0	56.4	−7.6	0.209
Co-located (%)	57.6	69.0	11.4	0.048
FTEs	156.0	107.0	−49.0	<0.001
Beds	63.9	49.7	−14.3	0.002
Total Salaries ($)	9,749,932	6,675,588	−3,074,344	<0.001

Note: Difference is "Ever Closed" minus "Never Closed." p-values are from two-sided Welch t-tests of equal means, allowing unequal variances across groups. Margins are reported as shares.

Over the course of my study period, 84 LTACH closed. Table 2 compares pre-period characteristics of surviving LTACHs to those that closed. I find that closing facilities are more likely to be smaller and less profitable leading up to the policy. The event study results in Figure 5b show that greater exposure to the policy is only significantly associated with closure in 2018 (0.019). This is potentially due to the compounding losses or in anticipation of the full implementation of the policy in the following years.

Table 3: LTACH Regression Results

	Dependent Variables										
	Price	Closure	Operating Margin	Medicare Margin	Commercial Margin	Total Discharges	Medicare Discharges	Commercial Discharges	FTEs	Salaries	Beds
Mean	36,233	84	0.042	0.062	0.099	493	317	135	156	9,702,046	64
Panel A: Difference-in-Differences											
$shock_{2013} \cdot interim$	49.8	0.005^{*}	0.002	0.005^{*}	0.002	−0.012	0.008	−0.011	0.002	−0.003	0.000
$shock_{2013} \cdot post$	$-1{,}049.5^{***}$	0.004	−0.002	-0.015^{***}	0.025^{***}	0.005	0.001	0.005	−0.006	−0.010	-0.009^{*}
year dummies											
2011	81.2	−0.002	0.005	0.001	0.034^{*}	−0.015	−0.009	-0.057^{**}	-0.047^{***}	-0.045^{***}	-0.030^{**}
2012	487.8^{***}	−0.002	0.009	0.002	0.046^{***}	0.000	0.011	−0.032	-0.022^{*}	-0.014^{**}	-0.021^{*}
2014	127.4	−0.007	0.003	−0.020	0.045	0.052	−0.049	0.086	−0.003	0.045^{*}	−0.010
2015	940.2^{***}	−0.001	−0.007	-0.022^{*}	0.012	0.052	−0.074	0.149^{***}	0.010	0.071^{***}	−0.012
2016	$1{,}603.5^{***}$	−0.021	-0.055^{***}	-0.032^{**}	-0.131^{***}	0.020	-0.116^{*}	0.136^{*}	−0.018	0.105^{**}	0.008
2017	$3{,}078.7^{***}$	0.005	-0.036^{**}	−0.022	-0.089^{**}	−0.015	-0.172^{***}	0.153^{**}	−0.021	0.107^{**}	0.009
2018	$3{,}933.9^{***}$	0.024	-0.055^{***}	-0.036^{**}	-0.108^{***}	−0.087	-0.307^{***}	0.160^{**}	−0.053	0.034	0.000
Panel B: Event Study											
$shock_{2013} \cdot year\ dummies$											
2011	−130.3	0.000	−0.004	−0.004	0.001	−0.012	0.010	−0.032	−0.005	0.007	−0.004
2012	−29.1	0.000	−0.003	0.001	−0.001	−0.022	0.000	−0.042	−0.009	0.004	−0.007
2014	−16.0	0.006	−0.001	0.004	0.002	−0.023	0.009	−0.040	−0.011	−0.001	-0.014^{***}
2015	9.8	0.004	0.000	0.004	0.002	−0.025	0.013	−0.030	0.006	0.001	0.006
2016	$-1{,}044.1^{***}$	0.003	-0.009^{*}	-0.013^{**}	0.017	−0.006	0.030^{**}	−0.028	−0.011	−0.001	−0.008
2017	$-1{,}058.8^{***}$	0.006	−0.001	-0.010^{*}	0.026^{***}	−0.017	0.023	−0.028	−0.006	−0.001	−0.008
2018	$-1{,}052.9^{***}$	0.019^{***}	0.003	-0.010^{*}	0.037^{***}	−0.034	−0.018	−0.033	−0.011	−0.027	-0.021^{***}
year dummies											
2011	707.2	−0.002	0.024	0.018	0.029	0.043	−0.055	0.092	−0.022	-0.076^{***}	−0.011
2012	628.2^{*}	−0.002	0.025^{*}	−0.001	0.049	0.102	0.012	0.164	0.020	-0.032^{*}	0.010
2014	443.9	−0.011	0.017	−0.016	0.044	0.101	−0.054	0.222^{**}	0.059	0.032^{**}	0.053^{**}
2015	$1{,}132.5^{***}$	0.002	0.002	−0.018	0.011	0.110	-0.099^{*}	0.241^{**}	−0.008	0.050^{**}	−0.042
2016	$1{,}821.2^{***}$	0.007	−0.013	−0.019	-0.089^{*}	0.014	-0.216^{***}	0.243^{**}	0.013	0.048	0.003
2017	$3{,}367.6^{***}$	0.020	−0.030	−0.021	-0.089^{*}	0.027	-0.238^{***}	0.261^{*}	−0.015	0.049	0.005
2018	$4{,}194.2^{***}$	−0.023	-0.069^{***}	−0.037	-0.156^{***}	0.037	-0.185^{*}	0.288^{**}	−0.024	0.098	0.058^{*}
N	3,040	3,040	2,736	2,732	2,726	2,680	2,731	2,734	2,706	2,736	2,736

Note: Statistical significance at the 10%(*), 5%(**), and 1%(***) levels. Discharges, FTEs, salaries, and beds are reported in log form; margins are reported as shares and closure as a binary indicator. Coefficients on $shock_{2013}$ interactions have been scaled by -1,000 for interpretability. Panel A reports the difference-in-differences specification and Panel B the event study specification. 2013 is the omitted reference year. Regressions of Price and Closure are based on the sample of all LTACHs. Subsequent regressions are conditional on surviving the entire study period.

I focus the remaining analysis on the 342 surviving LTACHs. Results are presented in columns 3-11 of Table 3. I begin with patient mix. I find no evidence of differential reductions in the number of Medicare discharges. Instead, my time trends show a common reductions following the policy (-0.116 to -0.307). Perhaps more interestingly, facilities were able to mitigate these reductions by substituting towards commerical discharges (0.136 to 0.160), resulting in no statistically significant differential or secular change in the total number of LTACH discharges. Given that surviving LTACHs appear to maintain their prior level of operations, it does not come as a surprise that I find no evidence that they adjusted either their variable costs (measured as FTEs and total salaries) or their fixed costs (measured as beds).

This translates to a both a secular and differential (-0.015) reduction in Medicare margin. This follows directly from the result that facilities did not differential reduce their Medicare discharges. I find a statistically significant increase in commercial margin associated with greater exposure to the site neutral policy as well as a reduction over time. This suggests that, while on average the newer commercial patients LTACHs are treating are less profitable, the facilities losing more on Medicare patients are able to get more from this payer. This could be for a variety of reasons. One story is that these facilities do a better job in screening for more profitable commercial patients. Alternatively, these facilities may negotiate higher commercial payments following the implemention of the policy. Unfortunately, I do not have data on these patients to further investiate this finding. These offsetting findings lead to no differential reduction in overall operating margin, but a statistically significant reduction over time.

6 Discussion and Conclusion

Site-neutral payment policies seek to align Medicare payment with patients' clinical needs rather than their site of care. I study the first such policy, which substantially reduced LTACH reimbursement for lower-acuity admissions beginning in 2016. Using simulated revenue changes based on pre-policy patients and post-policy rules, I find that LTACHs reduced admissions most in the DRGs facing the largest payment reductions, with an implied price elasticity of volume of 2.3. This decline was concentrated among site-neutral patients, with no detectable change in admissions of higher-acuity patients who remained eligible for the standard rate. These results suggest that the policy achieved its central aim of steering lower-acuity patients away from LTACHs without limiting access for those the facilities were intended to serve.

The magnitude of this response is also informative about provider behavior more broadly. My estimated elasticity is nearly double that of Gross et al. 2024 and four times that of Dafny 2005, consistent with LTACHs having greater discretion over admissions than acute care hospitals. This discretion may make prices a particularly effective tool for allocating patients across PAC providers.

At the facility level, the policy had meaningful financial consequences. Greater exposure was associated with larger declines in Medicare margins among surviving LTACHs, while the relationship between exposure and closure was weaker, emerging only in 2018. Surviving facilities offset declining Medicare volume by treating more non-Medicare patients, leaving total discharges unchanged. This substitution complicates the interpretation of the policy's savings. Because my commercial measure includes Medicare Advantage, some of this shift may reflect patients who remain Medicare beneficiaries. Moreover, the net effect on program spending depends on where patients no longer admitted to LTACHs received care and how much that care cost.

My results also offer indirect evidence on quality of care for the higher-acuity patients still treated at LTACHs. Although I do not directly measure quality, the volume and length of stay of non-site-neutral discharges were largely unchanged, and I find little evidence that more exposed facilities reduced staffing or salaries. Together, these results suggest that quality of care for these patients may not have changed despite lower Medicare margins. However, this conclusion should be interpreted with caution. Facility-level staffing measures may not capture changes in skill mix or in the care provided to specific patients, and my facility-level analysis is limited to LTACHs that survived the study period.

The effects on site-neutral patients are less clear and an important area for future research. Those who continued to be admitted had shorter stays, which could reflect either a change in treatment or a change in the composition of patients admitted. Patients who were no longer admitted may have received care in SNFs, remained longer in acute care hospitals, or been discharged to less intensive settings. Whether these changes affected mortality, readmissions, or total episode spending is central to evaluating the policy's welfare effects. If these patients achieved similar outcomes in less costly settings, then policy reduced wasteful spending. If instead they experienced worse outcomes, the savings would come at a cost to patients who may have benefited from LTACH care. Unfortunately, I am unable to evalaute changes for these individuals because they will only appear in my data conditional on receiving LTACH care.

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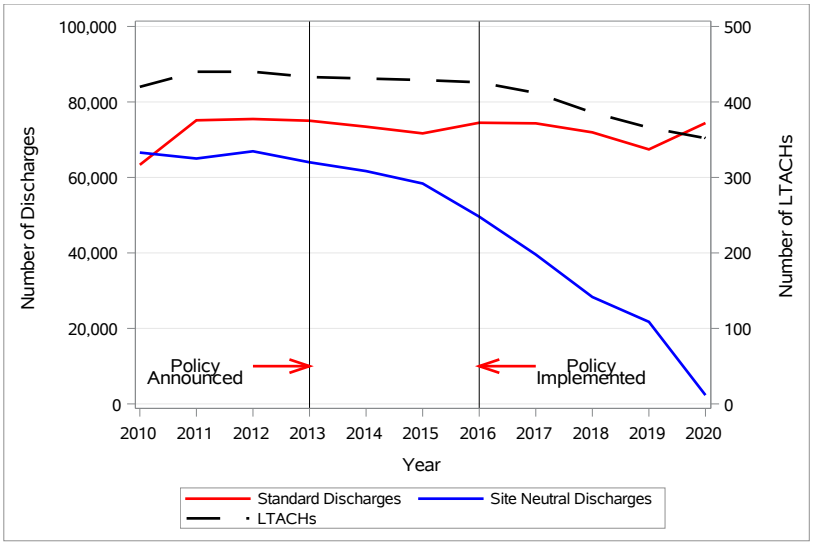
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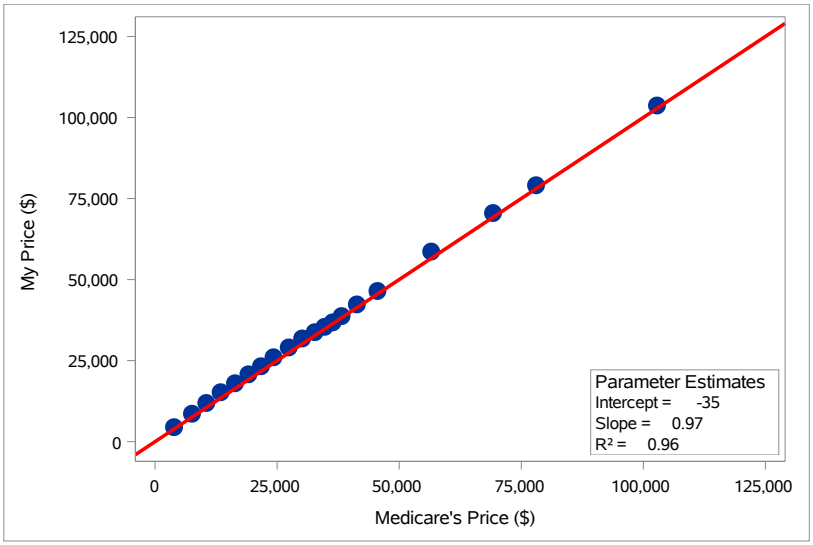
Appendix

Figure 6: Trend in LTACHs and LTACH Discharges



Note: LTACH trends based on CMS Cost Reports. Discharge trends based on Medicare fee-for-service discharges. Site neutral status for pre-2016 discharges determined by author's calculation of the policy definition. CMS waived the site neutral payment reform for the duration of the COVID-19 Public Health Emergency artificially reducing the number of site neutral discharges in 2020.

Figure 7: Binscatter of Realized Medicare Price to Predicted Medicare Price



Note: Based on Medicare fee-for-service LTACH discharges from fiscal years 2016-2018. Observations are binned into ventiles (5%) of realized Medicare price.

Table 4: DRG Regression Results

	Dependent Variables						
		All		Site Neutral		Non-Site Neutral	
	Price	Discharges	LOS	Discharges	LOS	Discharges	LOS
Mean	36,870	336	26	155	25	190	27
Panel A: Difference-in-Differences							
$shock_{2013} \cdot interim$	83.7^{***}	-0.001	-0.002^{***}	-0.004	-0.002^{**}	0.000	0.001
$shock_{2013} \cdot post$	$-1{,}020.8^{***}$	-0.062^{***}	-0.010^{***}	-0.044^{***}	-0.009^{**}	-0.007	-0.002^{*}
year dummies							
2011	207.9^{***}	-0.013	0.003	-0.027	0.003	-0.009	0.005
2012	615.4^{***}	-0.004	0.002	-0.007	0.001	-0.008	0.005
2014	-205.8^{**}	-0.016	0.007	0.008	0.015	-0.013	0.002
2015	211.4^{**}	-0.074	0.002	-0.060	0.006	-0.052	-0.004
2016	537.1	0.050	-0.025	-0.188^{*}	-0.068^{*}	-0.040	-0.021^{**}
2017	$1{,}822.4^{***}$	-0.058	-0.014	-0.422^{***}	-0.069^{*}	-0.064	-0.008
2018	$2{,}716.3^{***}$	-0.210	-0.014	-0.746^{***}	-0.094^{**}	-0.111	-0.005
Panel B: Event Study							
$shock_{2013} \cdot year\ dummies$							
2011	-58.8^{**}	0.004	0.001	0.002	-0.001	0.008	0.002
2012	-39.1	-0.001	0.001	-0.002	-0.001	0.001	0.002^{**}
2014	13.0	0.001	0.000	-0.007	-0.003^{*}	0.004	0.003^{***}
2015	97.3^{***}	-0.001	-0.002^{***}	0.000	-0.002	0.002	0.000
2016	$-1{,}009.2^{***}$	-0.045^{***}	-0.011^{***}	-0.032^{**}	-0.011^{**}	-0.007	-0.002
2017	-989.3^{***}	-0.058^{***}	-0.011^{***}	-0.039^{**}	-0.009^{*}	-0.004	0.000
2018	-885.4^{***}	-0.084^{***}	-0.011^{***}	-0.070^{***}	-0.014^{***}	-0.002	0.001
year dummies							
2011	493.2^{***}	-0.034	-0.003	-0.038	0.005	-0.048	-0.004
2012	806.6^{***}	-0.001	-0.002	0.002	0.004	-0.014	-0.006
2014	142.4	-0.028	-0.002	0.020	0.021	-0.030	-0.011^{*}
2015	151.4	-0.075	0.005	-0.077	0.002	-0.061	-0.003
2016	898.1^{**}	-0.039	-0.027	-0.264^{***}	-0.070^{*}	-0.039	-0.020^{*}
2017	$2{,}086.7^{***}$	-0.085	-0.015	-0.467^{***}	-0.079^{*}	-0.079	-0.015
2018	$2{,}471.2^{***}$	-0.111	-0.016	-0.634^{***}	-0.079^{**}	-0.134	-0.015
N	3,256	3,256	3,218	3,256	3,155	2,907	2,907

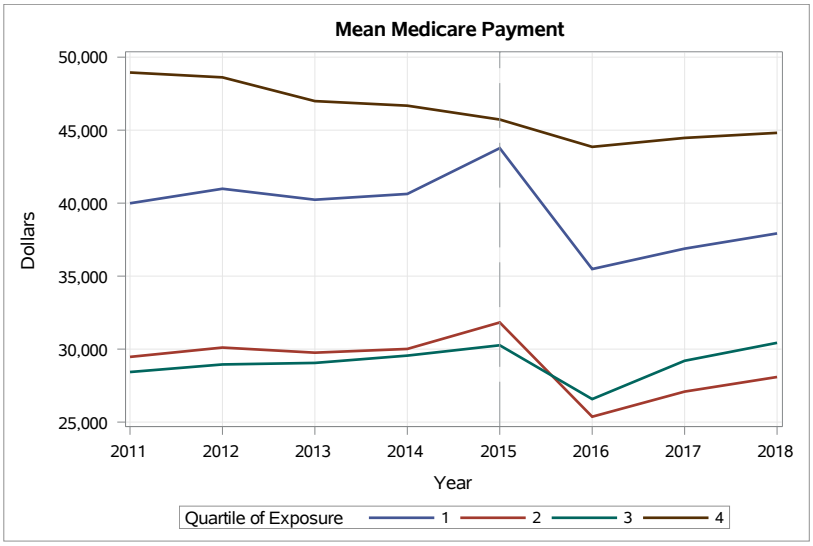
Note: Statistical significance at the 10%(*), 5%(**), and 1%(***) levels. Discharges and Length of Stay reported in log form. Coefficients on $shock_{2013}$ interactions have been scaled by -1,000 for interpretability. Panel A reports the difference-in-differences specification and Panel B the event study specification. 2013 is the omitted reference year.

6.1 DRG Weight Changes Over time

My primary model specifications include a vector of time-varying covariates for LTC-PPS and IPPS weights ($X_{d,t}$; $X_{h,t}$). These control for price changes attributable to changes in the underlying DRG weights during the annual recalibration process. Specifically, they are included to adjust for a reallocation of the LTC-PPS weights in 2015 that differentially increased the Medicare price of DRGs facing greater exposure to the site neutral policy. Figure 8 shows this in the time trends of Medicare price by quartile of exposure (Quartile 1 includes DRGs facing the greatest price

reductions and Quartile 4 includes the least). Table 5 presesnts regression results further verifying this. The first two columns show event study coefficients using Medicare price and LTC-PPS DRG weight as outcomes from equation (2) when dropping $X_{d,t}$. Columns 3 and 4 show the conversion factor for each year and the implied price difference from the weight change, respectively. While Column 4 overestimates the actual price difference, because not all discharges pass the SSO-threshold, it shows that deviations in the Medicare price estimates are fully attributable to changes in the weights.

Figure 8: Trend in Medicare Price by Quartile of Exposure



Note: LTACH sischarges are grouped into quartiles of pre-period exposure. Quartile 1 represents the set of DRGs facing the largest simulated price reductions following the the implementation of the site-neutral policy. Conversely, quartile 4 shows the set of DRGs facing the smallest simulated price reductions.

Table 5: Implied Price Difference by Year

	Medicare Price β	LTC-PPS Weight β	Conversion Factor ($)	Price Difference ($)
shock × 2011	-571.1^{***}	-0.0198^{***}	39,599.95	−784.07
shock × 2012	-499.9^{***}	-0.0163^{***}	40,222.05	−655.61
shock × 2013	-457.4^{***}	-0.0153^{***}	40,397.96	−618.08
shock × 2014	-360.4^{***}	-0.0129^{***}	39,808.74	−513.53
shock × 2015		*Reference year*		
shock × 2016	-976.2^{***}	0.0061^{***}	40,941.55	249.74
shock × 2017	-978.2^{***}	0.0039	41,641.49	162.40
shock × 2018	-918.9^{***}	0.0029	40,595.02	117.73

Note: Statistical significance at the 10%(*), 5%(**), and 1%(***) levels. Medicare Price and LTC-PPS Weight coefficients have been scaled by −1,000 for interpretability. Price Difference equals the LTC-PPS Weight β multiplied by the conversion factor. 2015 is the omitted reference year.

One concern is that this reflects regulatory capture. That is, LTACHs lobbied to increase Medicare price for DRGs they were going to lose the most on financially. The FY 2015 LTC-PPS Final Rule does not support this. One commenter requested upward adjustment for certain DRGs with lower weights relative to 2014, but respondents noted this is offset in other DRGs leaving aggregate payments budget neutral. Furthermore, it is explicitly stated that CMS is continuing to use the prior year's methodology (*79 FR 49854* n.d.). Nevertheless, if LTACH advocacy groups were able to negotiate behind the scenes, my results would be bias. The direction of this bias is not immediately clear. For example, assuming an upward-sloping supply curve, price increases would incentivize the admission of more patients. On the other hand, a higher price mechanically increases the difference between the LTC-PPS and the IPPS once the policy is implemented. This would encourage a larger post-period reduction. For completeness, I present the difference-in-differences results when dropping these controls in Table 6. These results are consistent with my preferred specification.

Table 6: DRG Difference-in-Differences Results: No Weight Covariates

	Dependent Variables						
		All		**Site Neutral**		**Non-Site Neutral**	
	Price	Discharges	LOS	Discharges	LOS	Discharges	LOS
Mean	36,870	336	26	155	25	190	27
$shock_{2013} \cdot interim$	283.3***	0.004	0.000	−0.002	−0.002	0.009	0.001
$shock_{2013} \cdot post$	−742.7***	−0.057***	−0.008***	−0.045***	−0.009	0.002	−0.001
year dummies							
2011	434.4**	−0.019	0.004*	−0.038	0.003	−0.009	0.005
2012	693.3***	−0.002	0.002	−0.007	0.001	−0.005	0.005
2014	−1,266.2***	−0.038	0.000	0.001	0.011	−0.049	−0.002
2015	−571.1	−0.094**	−0.003	−0.068	0.003	−0.081**	−0.007
2016	−1,347.3***	0.010	−0.036**	−0.190**	−0.072*	−0.108	−0.027***
2017	−121.6	−0.097	−0.025*	−0.424***	−0.073*	−0.132	−0.014*
2018	771.0	−0.254*	−0.026	−0.747***	−0.097**	−0.183	−0.010
N	3,256	3,256	3,218	3,256	3,155	2,907	2,907

Note: Statistical significance at the 10%(*), 5%(**), and 1%(***) levels. Discharges and Length of Stay reported in log form. Coefficients on $shock_{2013}$ interactions have been scaled by -1,000 for interpretability. 2013 is the omitted reference year.

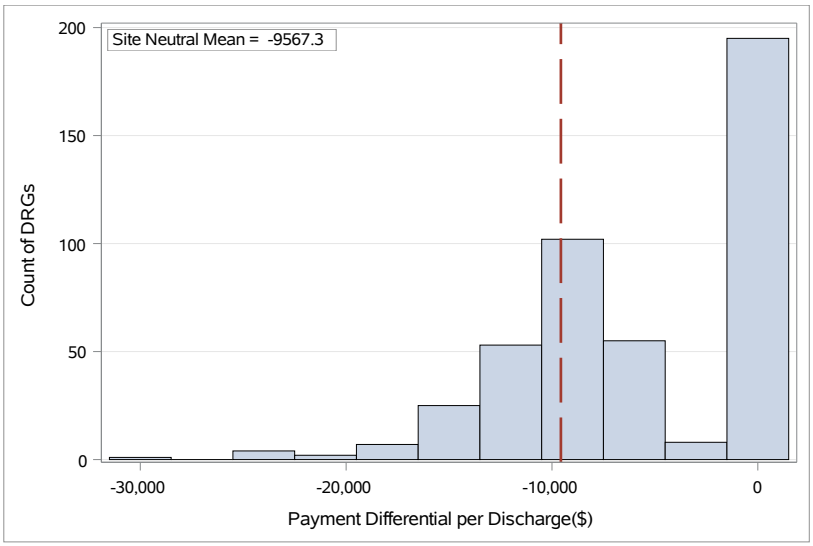
6.2 DRG-Site Neutral Specific Simulated Revenue Shocks

My DRG-level analysis uses an Intent-to-Treat design where my simulated revenue shock incorporates the distribution of site-neutral and non-site-neutral discharges within a DRG. I use this as my primary specification to guard against the concern that site-neutral assignment may be endogenous to the policy. For example, if LTACHs strategically shift discharges from site-neutral to non-site-neutral, my discharge results would be overestimated.

In my setting, this could occur before or during the LTACH hospitalization. Prior to the LTACH stay, transferring acute care hospitals could hold patients longer in the ICU so that they qualify for the full LTC-PPS rate. Alternatively, LTACHs could extend mechanical ventilation weaning to meet the 96 hour requirement. However, LTACH admissions account for around 1% of all Medicare acute care transfers and mechanical ventilation is measured coarsely in ICD-10 codes, suggesting either would require significant clinical justification. Given that the prices of non-site-neutral discharges are unchanged, it is potentially appealing to leverage this "unaffected" group in an analysis at the DRG-site-neutral status level. This allows me to estimate other causal parameters such as the average treatment effect for the treated (ATT) by leveraging within DRG changes across site neutral status. The distribution taking this approach is presented in Figure 9 followed by difference-in-differences estimates in Table 7. Results are consistent with my preferred

specification.

Figure 9: Distribution of DRG-Site Neutral Level Revenue Shock



Note: Shows the simulated per discharge price change under the site-neutral policy when calculated at the DRG-site-neutral level. Put differently, this dataset has two records per DRG. One for site-neutral discharges and one for non-site-neutral discharges. The former creates the distribution below zero, while the latter creates the mass point at zero. Based on 2013 discharges.

Table 7: DRG Regression Results, Separate Site Neutral and Non-Site Neutral Revenue Shocks

	Dependent Variables		
	Price	Discharges	LOS
Mean	36,475	342	26
$shock_{2013} \cdot interim$	23.1	0.000	-0.001^{*}
$shock_{2013} \cdot post$	$-1{,}083.3^{***}$	-0.051^{***}	-0.010^{***}
N	5,930	5,930	5,930

Note: Statistical significance at the 10%(*), 5%(**), and 1%(***) levels. Discharges and Length of Stay reported in log form. Coefficients on $shock_{2013}$ interactions have been scaled by -1,000 for interpretability. Panel A reports the difference-in-differences specification and Panel B the event study specification. Pre-period mean values are based on reference year 2013 discharges.

6.3 DRG Regression Results Excluding Psychiatric DRGs

Table 8: DRG Regression Results, Excluding Psychiatric DRGs

	Dependent Variables						
		All		**Site Neutral**		**Non-Site Neutral**	
	Price	Discharges	LOS	Discharges	LOS	Discharges	LOS
Mean	36,870	336	26	155	25	190	27
$shock_{2013} \cdot interim$	83.2***	−0.003	−0.002***	−0.005	−0.002*	0.000	0.001
$shock_{2013} \cdot post$	−1,023.7***	−0.061***	−0.010***	−0.043***	−0.009**	−0.007	−0.002*
N	3,114	3,114	3,114	3,114	3,114	3,048	2,910

Note: Statistical significance at the 10%(*), 5%(**), and 1%(***) levels. Discharges and Length of Stay reported in log form. Coefficients on $shock_{2013}$ interactions have been scaled by -1,000 for interpretability. Year dummies are included but not shown.